An Algebraic Reduction of the Five-Angle Oblique-Cutting Model and Explicit Cutting-Force Expressions

A technical note concerning Sec. 2.5 of Y. Altintas,
Manufacturing Automation, 2nd ed. (2012)

Doctoral course: Tool Engineering and Machining | Instructor: Dr. Peyman Mosaddegh

Scope and status. This note gives an algebraic reduction of the *maximum-shear-stress* formulation in Eqs. (2.48)–(2.52) and corresponding force expressions in Eqs. (2.53) and (2.59) of Ref. [1]. The result is exact *within that model and the stated angular branch*; it does not remove its mechanical assumptions or, by itself, establish higher experimental accuracy. No claim of literature priority is made.

1. Original five-angle problem

Let α_n (normal rake), β_a (mean friction angle), and i (cutting-edge inclination) be prescribed. The five unknown directions are the chip-flow angle η, resultant-force angles θ_n, θ_i, and shear-direction angles ϕ_n, ϕ_i. Altintas gives

$$\sin\theta_i = \sin\beta_a \sin\eta, \tag{2.48}$$

$$\tan(\theta_n + \alpha_n) = \tan\beta_a \cos\eta, \tag{2.49}$$

$$\tan\eta = \frac{\tan i \cos(\phi_n - \alpha_n) - \cos\alpha_n \tan\phi_i}{\sin\phi_n}, \tag{2.50}$$

$$\sin\phi_i = \sqrt{2}\sin\theta_i, \tag{2.51}$$

$$\cos(\phi_n + \theta_n) = \frac{\tan\theta_i}{\tan\phi_i}. \tag{2.52}$$

The final two equations follow from the maximum-shear-stress criterion: the resultant force makes a 45° angle with the shear direction and has no component transverse to the shear direction within the shear plane [1, pp. 22–23]. Equation (2.52) is to be interpreted by continuity when both tangents vanish.

2. Reduction to a single scalar equation

Define $s = \sin\beta_a \sin\eta$ and work on the conventional principal angular branch. Four angles follow immediately from η:

$$\theta_i = \arcsin s, \qquad \theta_n = \arctan(\tan\beta_a \cos\eta) - \alpha_n, \tag{6}$$

$$\phi_i = \arcsin(\sqrt{2}\, s), \qquad \phi_n = \arctan\left(\frac{1}{\sqrt{1-2s^2}}\right) - \theta_n. \tag{7}$$

In particular, $\tan\phi_i = \sqrt{2}\, s/\sqrt{1-2s^2}$. Rearrange Eq. (2.50) as

$$\tan i = \frac{\tan\eta \sin\phi_n + \cos\alpha_n \tan\phi_i}{\cos(\phi_n - \alpha_n)}. \tag{8}$$

Substitution of (6)–(7) into (8), followed by trigonometric simplification, yields a single equation for η:

$$\boxed{\tan i = \tan\eta \left(\sin\alpha_n + \frac{\cos\alpha_n}{\sqrt{1 - 2\sin^2\beta_a \sin^2\eta}}\right).} \tag{9}$$

A real principal-branch solution requires $1 - 2\sin^2\beta_a \sin^2\eta > 0$; other branches of the inverse trigonometric functions must be treated separately.

3. Quartic equation in the chip-flow direction

Introduce

$$t = \tan\eta, \quad p = \tan i, \quad A = \sin\alpha_n, \quad C = \cos\alpha_n, \quad K = \cos(2\beta_a). \tag{10}$$

Because $1 - 2\sin^2\beta_a \sin^2\eta = (1 + Kt^2)/(1 + t^2)$, Eq. (9) is equivalent, for the admissible branch, to

$$p - At = Ct\sqrt{\frac{1 + t^2}{1 + Kt^2}}. \tag{11}$$

Squaring and collecting powers of t gives

$$\boxed{\begin{aligned}&(A^2K - C^2)t^4 - 2pAKt^3 \\ &\quad + (A^2 + p^2K - C^2)t^2 - 2pAt + p^2 = 0.\end{aligned}} \tag{12}$$

Thus $\eta = \arctan t$ is algebraic of degree at most four. An expression in radicals exists in principle; the quartic representation is more compact and generally more useful computationally. **Squaring can introduce extraneous roots.** A candidate must satisfy the unsquared Eq. (11), the geometric domain, and the original five equations. The physically intended solution corresponds to the angular branch consistent with the cutting geometry.

4. Cutting forces without five-angle iteration

For uncut chip thickness h, nominal chip width b, and mean shear-plane flow stress τ_s, the book's Eq. (2.53) is

$$F_c = \frac{\tau_s bh}{\left[\cos(\theta_n + \phi_n)\cos\theta_i \cos\phi_i + \sin\theta_i \sin\phi_i\right]\cos i \sin\phi_n}. \tag{13}$$

The maximum-shear-stress construction makes the bracketed shear-force projection equal to $\cos 45^\circ = 1/\sqrt{2}$, so

$$F_c = \frac{\sqrt{2}\,\tau_s bh}{\cos i \sin\phi_n}. \tag{14}$$

For concise expressions define the four dimensionless quantities

$$\begin{aligned} X &= \cos\beta_a \cos\alpha_n + \sin\beta_a \cos\eta \sin\alpha_n, \\ Y &= \sin\beta_a \cos\eta \cos\alpha_n - \cos\beta_a \sin\alpha_n, \\ Z &= \sin\beta_a \sin\eta, \qquad Q = \sqrt{1 - 2Z^2}. \end{aligned} \tag{15}$$

From Eqs. (2.48)–(2.49), $\cos\theta_i \cos\theta_n = X$, $\cos\theta_i \sin\theta_n = Y$, and $\sin\theta_i = Z$. Further,

$$\sin\phi_n = \frac{X - QY}{\sqrt{2}(1 - Z^2)}. \tag{16}$$

Substituting into Eq. (14) gives the *shearing-only* resultant force,

$$\boxed{F_c = \frac{2\tau_s bh(1 - Z^2)}{\cos i\,(X - QY)}.} \tag{17}$$

All quantities on the right depend only on $\alpha_n, \beta_a, i, \tau_s, bh$ and the admissible algebraic root t of (12).

4.1. Resolved cutting-force components

The projections specified by Altintas's Eq. (2.59) are

$$\begin{aligned} F_{tc} &= F_c(\cos\theta_i \cos\theta_n \cos i + \sin\theta_i \sin i), \\ F_{fc} &= F_c \cos\theta_i \sin\theta_n, \\ F_{rc} &= F_c(\sin\theta_i \cos i - \cos\theta_i \cos\theta_n \sin i). \end{aligned}$$

Using Eqs. (15)–(17) leads to the compact expressions

$$\boxed{F_{tc} = \frac{2\tau_s bh(1-Z^2)(X\cos i + Z\sin i)}{\cos i\,(X - QY)}} \tag{18a}$$

$$\boxed{F_{fc} = \frac{2\tau_s bh(1-Z^2)Y}{\cos i\,(X - QY)}} \tag{18b}$$

$$\boxed{F_{rc} = \frac{2\tau_s bh(1-Z^2)(Z\cos i - X\sin i)}{\cos i\,(X - QY)}} \tag{18c}$$

The corresponding shearing-force coefficients K_{tc}, K_{fc}, K_{rc} follow by dividing these expressions by bh, in agreement with the interpretation of Eq. (2.61). Where $F_{tc} \neq 0$, the force ratios are

$$\boxed{\frac{F_{fc}}{F_{tc}} = \frac{Y}{X\cos i + Z\sin i}, \qquad \frac{F_{rc}}{F_{tc}} = \frac{Z\cos i - X\sin i}{X\cos i + Z\sin i}.} \tag{19}$$

For a given set of model angles, these ratios are independent of the magnitude of $\tau_s bh$. The force components here follow the directions/sign conventions of Ref. [1]. The edge/ploughing contributions discussed around Eqs. (2.60)–(2.61) must be added separately when modeling the measured total forces.

5. Numerical consistency and limiting cases

For $\alpha_n = 10°$, $\beta_a = 30°$, $i = 15°$ on the stated branch:

Direction or force	Symbol	Value
Chip-flow angle	η	12.88921864°
Resultant-force elevation	θ_i	6.40371377°
Resultant-force normal-plane angle	θ_n	19.37093458°
Shear-direction inclination	ϕ_i	9.07527608°
Normal shear angle	ϕ_n	25.98993473°
Resultant force / $\tau_s bh$	$F_c/(\tau_s bh)$	3.34107110
Tangential component / $\tau_s bh$	$F_{tc}/(\tau_s bh)$	3.12198764
Feed component / $\tau_s bh$	$F_{fc}/(\tau_s bh)$	1.10126092
Radial component / $\tau_s bh$	$F_{rc}/(\tau_s bh)$	−0.45074788

Direct evaluation of Eqs. (2.48)–(2.52) gives maximum absolute residual 1.4×10^{-17}; the force components from Eqs. (18) and the unsimplified Eq. (2.59) differ by at most 4.5×10^{-16} after normalization by $\tau_s bh$. A reproducible Python script accompanies this note. Across 500 seeded test points with $\alpha_n \in [-10°, 20°]$, $\beta_a \in [5°, 40°]$, $i \in [-25°, 25°]$, the largest absolute residual of the five equations was 3.0×10^{-14}, and the maximum difference of normalized force components was 5.6×10^{-13}.

5.1. Two useful limits

In the orthogonal limit $i = 0$, the principal solution gives

$$\eta = \theta_i = \phi_i = 0, \quad \theta_n = \beta_a - \alpha_n, \quad \phi_n = 45^\circ - \beta_a + \alpha_n. \tag{20}$$

This reproduces the maximum-shear-stress orthogonal-cutting relation in the book. In the *frictionless limit* $\beta_a \to 0$, Eq. (9) reduces to

$$\boxed{\tan\eta = \frac{\tan i}{\sin\alpha_n + \cos\alpha_n}}. \tag{21}$$

Therefore the frequently used empirical relation $\eta = i$ is *not* a general mathematical consequence of zero friction. For ordinary principal angles it follows in particular when $\alpha_n = 0$ (or trivially when $i = 0$). Equation (2.52) at exactly $\beta_a = 0$ is interpreted by taking its regular limit.

6. Interpretation, limitations, and possible follow-up

1. **What is obtained.** Equations (9)–(12) reduce the five-angle maximum-shear-stress system to a quartic in $\tan\eta$; Eqs. (17)–(19) then give force components and force ratios as algebraic functions of the inputs. This removes the need to iterate all five angles in this formulation.
2. **What is not changed.** The assumed shear-plane geometry, the maximum-shear-stress criterion, a mean friction angle, and the effective shear flow stress are unchanged. The result is not an experimental force correction and does not cover edge forces by itself.
3. **Relation to other approaches.** The book also treats an alternative minimum-energy criterion, an Armarego–Whitfield empirical model with three geometric equations, and practical approximations such as $\eta = i$. The present reduction applies to the *maximum-shear-stress* branch only; it is not a solution to the minimum-energy stationarity system.
4. **Possible assessment.** It would be helpful to compare robust root selection, conditioning, numerical cost, and predictions against the original iterative method, and to evaluate against observed chip-flow angles and experimentally measured shearing-force components after edge-force corrections.

Attribution and disclosure

These manipulations were developed with the assistance of generative AI and then checked by numerical substitution against the cited textbook equations. The checks establish *internal mathematical consistency*, not originality, proof of global branch uniqueness, or experimental validity. I would particularly welcome a critical assessment of the reduction, any relevant prior derivations, and the domain in which the stated angular branch is appropriate.

Reference [1] Y. Altintas, *Manufacturing Automation: Metal Cutting Mechanics, Machine Tool Vibrations, and CNC Design*, 2nd ed., Cambridge University Press, 2012, Sec. 2.5, pp. 19–26.